Renewable Energy 248 (2025) 123099

Contents lists available at ScienceDirect

Renewable Energy

journal homepage: www.elsevier.com/locate/renene

Experimental optimization of stacked solar PV panels: Strategic positioning and reduced footprint for power maximization

Gurukarthik Babu Balachandran [a,*], Muthu Eshwaran Ramachandran [b], M. Palpandian [c], Prince Winston David [a]

[a] Department of Electrical and Electronics Engineering, Kamaraj College of Engineering and Technology, K.Vellakulam, Near Virudhunagar - 625701, Tamil Nadu, India
[b] Department of Artificial Intelligence and Datascience, Ramco Institute of Technology, Rajapalayam, 626117, Tamil Nadu, India
[c] Department of Electrical Engineering, Annasaheb Dange College of Engineering and Technology, Ashta, Maharashtra, 416 301, India

ARTICLE INFO

Keywords:
Building-integrated photovoltaic (BIPV)
Photovoltaic system optimization
Interconnection schemes
Power output maximization
Solar panel positioning
Stacking

ABSTRACT

Solar energy is a clean, non-polluting energy source. Photovoltaic (PV) systems are expected to play a crucial role in future electricity generation. This study explores innovative strategies to maximize PV panel output by optimizing panel positioning and reducing surface area. Through controlled experiments, different PV panel positions were analyzed to determine their impact on power generation. The study also examined the trade-off between reducing panel area and maintaining energy efficiency. Results indicate that both panel positioning and area reduction significantly influence energy generation. The highest output power was achieved with a flat-position PV array at a 93 cm variance using Series-Parallel (Se-P) interconnection (156 W), followed by 77 cm (106 W) and 62 cm (135 W) configurations. These findings provide valuable insights for designing efficient PV systems, particularly for space-constrained environments. Unlike previous studies, this work systematically investigates the synergy between panel positioning and area optimization, offering a novel approach to enhancing PV system performance. The results highlight the practical applications of optimized PV configurations in urban areas and regions with limited land availability, contributing to the advancement of sustainable solar energy solutions.

Nomenclature

Photovoltaic	PV
Flat Positiond Photovoltaic	FP PV
V Positiond Photovoltaic	VP PV
Inverted V Positiond Photovoltaic	IVPPV
Series-Parallel	Se-P
Total-Cross-Tied	TCT
2 step Stair Case	2SS
Fill Factor	FF

1. Introduction

Due to the exhaustion of fossil fuels, the demand for renewable energy sources has increased gradually over the years. Among the possible renewable energy sources, solar PV systems are considered the most promising and feasible sources that offer clean and sustainable energy. The solar PV system has gained attention because of the development of power transformation technology and the reduction in the cost of PV cells [1,2]. However, the performance of PV systems depends on environmental conditions. The output power of the PV system depends on incident irradiation, temperature, partial shading and soiling. In these, partial shading caused by the buildings, trees, clouds and dust or snow significantly impacts the reduction of output power [3]. Under these partial shading conditions, some of the series-connected panels may not receive uniform irradiation, which causes mismatch losses between the array rows. Due to this, mismatch current flow between the shaded and unshaded panel results in multiple power peaks and leads to a reduction in output power. This makes it difficult for the maximum power point tracking controller to track the global power peak. The extraction of maximum power from the partially shaded PV array depends on the array configurations, shade patterns, array arrangements, position and area of shade [4,5]. This study will focus on the array arrangements and configurations for further analysis.

Conventional PV array configurations like series, parallel, series-

* Corresponding author.
E-mail addresses: mspsbguru@gmail.com, gurukarthikbabueee@kamarajengg.edu.in (G.B. Balachandran).

https://doi.org/10.1016/j.renene.2025.123099
Received 9 January 2025; Received in revised form 14 March 2025; Accepted 8 April 2025
Available online 9 April 2025

parallel (Se-P), bridge link, honeycomb and total-cross-tied (TCT) are employed to enhance the output power [6]. Similarly, hybrid configurations like series parallel total-cross-tied, bridge link total-cross-tied, honeycomb total-cross-tied, and bridge link honeycomb configurations are utilized to improve power generation. Compared to conventional and hybrid configurations, the TCT configuration delivered better results by reducing the mismatch power loss. However, the TCT fails to extract the maximum power, if the series of connected panels are shaded [7]. To overcome the drawbacks of TCT configuration, static and dynamic PV array reconfiguration techniques are anticipated.

The static reconfiguration technique is a predefined technique in which the physical location of the panels is changed according to the puzzle pattern or mathematical approach. The electrical connections between the panels remain unaltered [8]. In Ref. [9], proposed the integration of the Sudoku puzzle pattern within a TCT PV array. The Sudoku-based approach outperforms the conventional Se-P configuration in terms of maximum power. Further, the dispersion of shade is improved by the optimal Sudoku [10] and improved Sudoku [11]. In Ref. [12], the magic square puzzle pattern for TCT, Series parallel total-cross-tied, bridge link total cross and bridge link honeycomb configuration. Among the configurations reconfigured TCT produced superior results than the conventional TCT. The Ken-Ken mathematical puzzle pattern enhances the output power by optimally placing the panels within the array [13]. The dominant square approach alters the physical position of the panels strategically in a PV array [14]. The competence square technique employs a logic-based placement strategy to fix the optimal location of the panels in the PV array [15]. In Ref. [16], proposed staircase static techniques like 1 step staircase, 2 step staircase (2SS), 3 step staircase and 4 step staircases for 5 x 5 PV array. The author concluded that the 2SS produced superior results in terms of maximum power.

In the dynamic reconfiguration technique, the physical positions of panels are adjusted dynamically according to the varying partial shading conditions. This approach involves real-time data, optimization algorithms, switches and sensors [17]. In Ref. [18], proposed the one-step reconfiguration for enhancing the output power of a 4 x 4 PV array. In Ref. [19], the logic-based dynamic reconfiguration to have the identical row current of the TCT array under partial shading conditions. This algorithm measures the short-circuit current of fixed and adaptive parts by isolating the load. In Ref. [20], proposed the cost-effective neuro-fuzzy algorithm to improve the output power of the PV array. The adaptive evolutionary jellyfish search algorithm optimally places the panels within the array. This approach offers superior results compared to other metaheuristic algorithms [21]. In the current injection technique, the mismatch current is injected to enhance the output power of the array. This technique involves dynamic reconfiguration to reduce the mismatch current between the rows of the array. Additionally, the compensation current is injected into the rows based on the existence of a mismatch current. The dynamic reconfiguration techniques suffer due to the requirement of real-time data, complex algorithms, switches and sensors [22]. However, the static reconfiguration technique avoids the shortcomings of the dynamic reconfiguration technique and is found to be cost-effective. Further, this study considers the conventional Se-P, TCT and 2SS [16] for further analysis.

The PV orientation plays a significant role in the expansion of the PV system. Since the land requirement of PV depends on the orientation structure. It is essential to locate the panel with a reduced land scape. The PV panels should be mounted with an optimal tilt angle to acquire the highest yearly average incident irradiation [23]. The conventional PV array is ground mounted, in which the landscape depends on the size of the array [24]. The building integrated PV array may reduce the land requirements along with cost and environmental benefits [25]. Similarly, the solar trees were employed to capture the sunlight throughout the day along with reduced land requirements. In solar trees, the panels were attached to the tree-like structure as leaves. The power generation from the solar tree depends on the tilt angle, inverter ratio and row spacing [26]. The building-integrated PV array and solar tree generate more output power because of better cooling of wind in comparison to the ground-mounted PV array [27]. Hence, the variance of the panel plays a vital role in dissipating the heat generated by the panel.

From the literature, it is observed that the power generation of the array depends on the incident irradiation on the panel which varies with the orientation structure of the array. To capture the sunlight throughout the day, this study proposes the V-positioned PV (VP PV) and inverted V-positioned (IVP PV) array to enhance power generation. The performance of the proposed VP PV and IVP PV is compared with the conventional flat position (FP) PV array. The performance of FP, VP and IVP PV arrays are tested under different configurations like Se-P, TCT and 2SS [16]. Further, the PV array under different orientations and configurations is also tested for different variances of 62 cm, 77 cm and 92 cm respectively.

Enterprises buying carbon emission rights face price fluctuations and must time their purchases wisely. This study uses a mathematical model to help businesses decide the best time to buy, considering demand and inventory. A new technique is introduced to solve this problem, offering practical guidance for carbon asset management [33]. Economic growth depends on strong infrastructure and sustainable practices. Special economic zones (SEZs) and energy projects attract investments and promote stability. This research shows how supply chain management and glocalization help economies become more resilient by ensuring sustainability and balancing global and local needs [34].

Global installed capacity of solar PV has grown continuously since 2000. In 2023, the global installed capacity of solar PV energy will reach 1177 GW. This growth in the solar PV market reflects a global shift towards renewable and sustainable energy technologies. China and the United States lead the global PV market, with 307 and 122 GW of installed solar PV capacity, respectively. On the other hand, Chile and Honduras had the highest share of PV energy mix in total energy produced in 2023 [35].

The importance of PV solar energy fields is paramount, especially given its role as an optimal complement to both renewable and conventional energy sources within hybrid energy systems. These systems are widely deployed globally due to their robustness and reliability in energy production from various sources, including PV/grid, PV/wind, PV/diesel, PV/concentrated solar power (CSP), PV/wind/diesel, and PV/wind/battery configurations. Nassar and Alsadi (2019) assessed the solar energy potential in the Gaza Strip, Palestine, emphasizing its feasibility for sustainable energy generation in the region. Their study highlighted the importance of solar PV deployment for energy security and reducing dependence on conventional energy sources [36].

Nassar et al. (2025) investigated the role of hybrid renewable energy systems in mitigating power shortages in public electricity grids. Their research focused on economic, environmental, and technical optimization, demonstrating that hybrid PV-wind systems can significantly improve grid reliability and energy affordability [37]. To enhance energy storage solutions, Nassar et al. (2021) proposed an optimized pumped hydroelectric storage-integrated PV/Wind system. Their dynamic analysis identified key factors influencing system efficiency, suggesting that pumped hydro storage can effectively support hybrid renewable energy generation [38].

In the context of sustainability in Palestine, Nassar et al. (2024) explored renewable energy potential and proposed strategic policies to enhance long-term energy security. Their findings provided policy recommendations for integrating renewable sources into national energy planning [39]. El-Khozondar et al. (2023) designed a standalone hybrid PV/Wind/Diesel generator system to support power supply in COVID-19 quarantine centers. Their study demonstrated the feasibility of decentralized renewable energy solutions for emergency response applications [40].

Further research by Nassar et al. (2024) introduced a standalone utility-scale pumped hydroelectric storage system powered by a hybrid PV/Wind setup. Their study underscored the importance of reliable energy storage in facilitating the large-scale adoption of renewables [41]. Table 1a which clearly notates the survey of the previously published relevant works.

1.1. Mathematical modeling

1.1.1. PV modeling

The real power (Ppv) of the PV panel under real operation and climatic conditions (solar radiation Ht and ambient air temperature T∞) is estimated [43] by (Equation (1)):

$$P_{pv} = P_{stc} \times DF_{pv} \times (Hg/H_{stc}) \times [1+\beta_p (T_{cell}\text{-}T_{stc})] \quad (1)$$

Where:

P_{stc} is the nominal PV power at the Standard Test Condition STC

β_p is the power coefficient (%/°C); T_{stc} and H_{stc} are the standard test conditions of temperature (25 °C) and solar radiation (1000W/m^2), respectively.

The cell's temperature (Tcell) [44] is expressed as (Equation (2)):

$$T_{cell} = T_\infty + 0.078H_g \quad (2)$$

where:

T_∞ is the ambient air temperature (°C).

H_g is the global inclined solar irradiance (W/m^2).

The mathematical representation of PV system performance using the standard single-diode has been deduced from the survey as represented in the Technical and Economic Feasibility of Utility-Scale Solar Energy Conversion Systems in Saudi Arabia (Equation (3)):

$$I = Iph \text{ - } IO \left(e^{(q(V+IRs)/\text{nkT})} - 1\right) - \left((V+IRs)/Rsh\right) \quad (3)$$

where:

I - Output current.

I_{ph} - Photocurrent.

I_0 - Diode saturation current

q - Electron charge.

V - Output voltage.

R_s and R_{sh} - Series and Shunt resistances

n - ideality factor

k - Boltzmann's constant.

T - temperature in Kelvin.

1.1.2. Solar irradiance model

An analytical model for solar irradiance estimation, such as the **Liu & Jordan transposition model** for tilted surfaces has been estimated based on the solar irradiance on solar fields from an analytical approach and experimental results (Equation (4)):

$$G_{T=} G_bR_b + G_dR_d + G_rR_r \quad (4)$$

where:

Table 1(a)
Comparison of the Literature survey.

Reference	Type of Panel	Arrangement	Novelty of the Work	Major Results	Inferences	Research Gap
[28]	Photovoltaic (PV) and Concentrated Solar Power (CSP)	Evaluated land use for both PV and CSP systems near the Tropic of Cancer region	Comparison of land-use efficiency between PV and CSP technologies near the Tropic of Cancer	•PV systems tend to have better land-use efficiency compared to CSP systems in the studied regions. •CSP land use: 2.5 ha/MW, PV land use: 1.5 ha/MW, Global Horizontal Irradiance (GHI) influences land use	The study suggests PV systems as more competitive in terms of land use near the Tropic of Cancer.	•Further research needed on optimizing land use in hybrid PV-CSP systems •Analysis required on other environmental factors (e.g., water use) for CSP and PV systems.
[29]	Perovskite/Silicon Tandem Cell	Mechanically stacked, two-terminal tandem cell configuration	Developed a graphene-based perovskite/silicon tandem solar cell achieving an efficiency of over 26 %.	Demonstrated efficiency improvement (26 %) in tandem structure using graphene, surpassing traditional silicon-only cells.	Higher efficiency can be achieved using stacked structures with novel materials like graphene.	The long-term stability of graphene-based perovskite tandem cells needs further investigation.
[30]	Shotcrete PV racking system	Studied direct and low-concentration cases of PV racking systems	Introduces a shotcrete-based racking system to improve geographic adaptability of PV installations.	•Direct PV case potential: 1140 GW, low-concentration case: 1350 GW •Shotcrete PV racking shows promise for use in a wider range of geographic regions with better structural integrity.	Suggests that shotcrete-based systems can enhance the geographic flexibility of PV installations.	•The environmental impact of shotcrete racking systems and cost-effectiveness need to be studied in more detail. •Further research needed on technical and economic feasibility in different climates
[31]	Rooftop PV systems	Rooftop installations across multiple identical systems analyzed	Analyzed performance variability of rooftop PV systems, focusing on yield prediction and real-world performance.	•Performance variability up to 10 %, significant deficit in 15 % of systems. •Performance variability was significant, leading to implications for more accurate yield predictions for rooftop PVs.	The study highlights the importance of considering system variability in yield predictions for rooftop PV systems.	•Further research is needed on improving yield prediction models considering environmental and technical factors. •Lack of investigation into the root causes of underperformance in certain rooftop systems
[32]	Bifacial PV panels	Agrivoltaic systems combining bifacial PV with olive tree cultivation	Explores the integration of bifacial PV panels in agrivoltaic systems to enhance land use efficiency.	•Bifacial PV panels showed a 15 % increase in energy production in combination with olive trees. •Bifacial PV combined with olive trees can increase energy production while maintaining agricultural productivity.	Shows the potential for combining energy generation with agriculture to maximize land use in agrivoltaic systems.	•Long-term impacts on crop yield and panel performance in agrivoltaic systems need further exploration. •optimal crops and PV configurations for agrivoltaic systems in diverse climates is needed

G_T - total irradiance on a tilted plane.
G_b, G_d, and G_r - beam, diffuse, and reflected radiation.
R_b, R_d, and R_r - transposition factors.

1.1.3. Economic and environmental modeling

The **Levelized Cost of Energy (LCOE)** equation has been represented as (Equation (5)):

$$LCOE = \frac{C + \sum_{t=1}^{N} (O_t + M_t)/(1+r)^t}{\sum_{t=1}^{N} (E_t)/(1+r)^t} \tag{5}$$

The **Carbon Footprint Analysis** has been done using **Life Cycle Assessment (LCA)** methodologies in reference with the energy savings strategy for the residential sector in Libya and its impacts on the global environment and national economy (Equation (6)):

$$CO_{2\ savings} = E_{total} \times EF \tag{6}$$

where:
E_{total} - Total energy generated over the PV panel's lifetime.
EFEF - Grid emission factor (kg CO_2/kWh).

1.2. Digest of research gap

1. **Limited research on stacking arrangements:** While various PV configurations have been explored, there is a gap in experimental studies focusing on the effectiveness of stacked PV panels for power maximization.
2. **Inadequate exploration of footprint reduction:** Few studies have systematically addressed how reducing the surface area of PV systems, through strategic positioning, can optimize power output.
3. **Lack of comprehensive experimental validation:** Most literature lacks rigorous experimental data on the real-world performance of stacked PV systems in spatially constrained environments.
4. **Insufficient analysis of interconnection impacts:** The influence of interconnection strategies between stacked PV panels on overall energy yield remains underexplored.
5. **Need for spatial optimization models:** Current studies do not fully investigate models that integrate both strategic placement and stacking to enhance efficiency in limited spaces.

1.3. Statement of Novelty

1. Stacking Arrangement as a Key Strategy

The work also explores the stacking arrangement of PV panels as a critical method for improving power output. By utilizing a vertical stacking approach, the study aims to increase the efficiency of energy capture in space-constrained environments.

2. Experimental Configurations and Strategic Adjustments

Through systematic testing of various configurations and detailed experimentation, the research highlights how strategic adjustments in panel orientation and footprint reduction can significantly enhance the power output of PV systems. This process involves evaluating different positioning scenarios to identify optimal configurations.

3. Insights into Maximizing Solar Energy Efficiency

The findings offer novel insights into maximizing solar energy efficiency, providing a new perspective on how spatial considerations can influence overall energy yield. The research lays the groundwork for further investigations into the interplay between positioning, stacking, and power generation.

4. Practical Solutions for Space-Limited Deployments

Results demonstrate that a careful evaluation of panel placement not only improves energy yield but also provides practical solutions for deploying PV systems in space-limited environments. This is particularly relevant for urban areas or regions with limited land availability.

5. Addressing Contemporary Renewable Energy Challenges

This research addresses key challenges in renewable energy utilization by offering effective strategies for optimizing solar power generation in spatially constrained areas. It paves the way for future advancements in PV system design and deployment, contributing to more sustainable energy solutions.

1.4. Objective of the study

- Novel design scheme to reposition the PV array configuration to enhance the maximum power and optimize the landscape.
- The performance of FP PV, VP PV and IVP PV array with different interconnections schemes (Se-P, TCT, 2SS [16]) are investigated mounted at a variance of 62 cm, 77 cm and 92 cm respectively.
- The performance of the PV array at different positions and interconnections is analyzed in terms of output power and FF.

1.5. Dissection of the work

The remaining section of this paper is written as follows; Section 2 describes the experimentation of the proposed work with different positions of PV array for various interconnection schemes mounted at different variances. Section 3 presents the comprehensive analysis of the proposed work followed by the conclusion in Section 4.

2. Description of proposed work

This study investigates the enhancement of power generation on PV arrays positioned at different positions, different interconnections and different variances. The experimental set-up consists of 12 numbers of 10 W PV panels connected to form a 3 x 4 array of Se-P, TCT and 2SS [16]. The PV panels are positioned at three different positions namely FP PV, VP PV and IVP PV as shown in Fig. 1. In FP PV, the PV panels are placed horizontally at 180°. The FP PV have the advantage of capturing 90 % of the solar irradiation. In VP PV and IVP P, the PV panels are positioned in letter V and inverted V position with a tilt angle of 45°. For VP PV and IVP PV, the incident solar irradiation on panels depends on the position of the sun. The interconnections like Se-P, TCT and 2SS [16] were considered for the analysis.

The experimental setup for placing the PV panels at a different variance is obtained as follows. The PV panels are mounted on the four-handed variant, which is supported by ten-foot iron rods. The thickness of the iron rod is about half an inch. A half-inch hole is punched on every ten feet of iron rods to place the PV panels at a different variance. At the centre bottom, eight-feet iron rods are positioned horizontally to place the PV panels. The electrical specifications of PV panels and measuring instruments utilized for the study are presented in Table 1(b). Thus, the mounting set-up provides the provision to place the PV panels at a different variance of 62 cm, 77 cm and 92 cm. The power generation of PV panels is assessed for different interconnections such as Se-P, TCT, 2SS [16], and different positions like FP PV, VP PV and IVP PV at variance of 62 cm, 77 cm and 92 cm as shown in Fig. 2.

The experimentation verification is conducted at Kamaraj College of Engineering and Technology, Virudhunagar (Latitude 9.67275° North; Longitude 77.96475° East). The three positions of the PV array were exposed to solar irradiation for different interconnection schemes and variances. The electrical quantities of the array were recorded from 09.00 a.m. to 06.00 p.m. with a time interval of 30 min. The incident

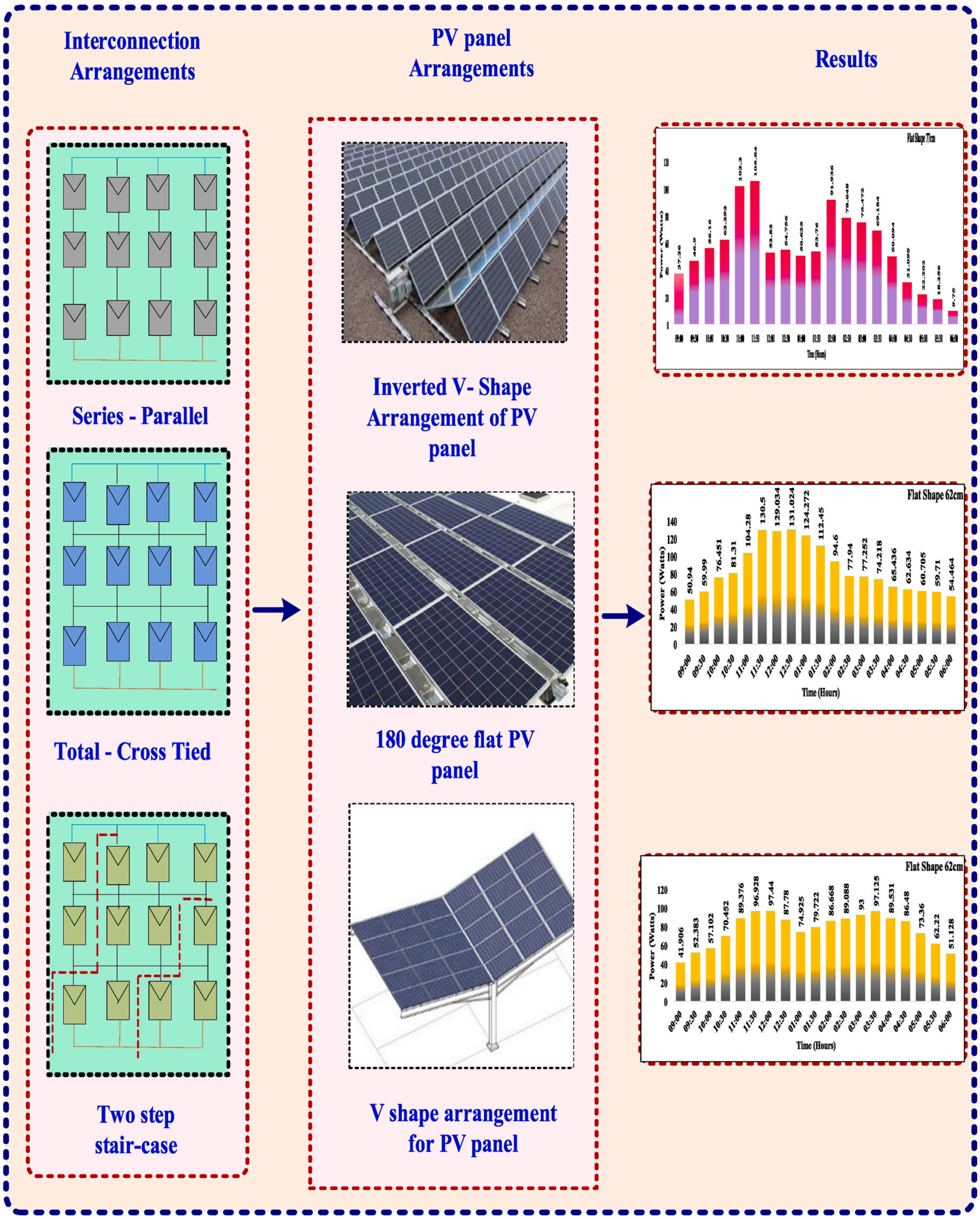

Fig. 1. Block diagram of the proposed work.

irradiation on the PV panels is measured by a pyranometer. The electrical quantities of the array are measured by using the voltmeter and ammeter.

2.1. Assumptions

1. **Homogeneity of PV Panel Performance:** It is assumed that all PV panels used in the experiments are of the same type and model, ensuring consistent performance characteristics, efficiency ratings, and output across all tests.
2. **Controlled Environmental Conditions:** The experiments assume that environmental conditions, such as temperature, humidity, and solar irradiance, are controlled or monitored throughout the study to minimize their influence on the power output of the PV panels.
3. **Linear Relationship between Panel Position and Power Output:** The study assumes that the relationship between panel positioning and power output follows a predictable linear or quadratic trend, allowing for effective modeling and optimization using statistical methods.

Table 1(b)
Solar PV panel Characteristics.

Parameters	Electrical Specification
Maximum Power (P_m)	10W
Maximum Power Voltage (V_m)	20V
Maximum Power Current (I_m)	0.50A
Short Circuit Current (I_{sc})	0.60A
Short Circuit Voltage (V_{oc})	24.8V
Maximum System Voltage	600V
Ammeter (MC Meter)	(0-20) A
Voltmeter (MC Meter)	(0–300) V
Pyranometer	(0–2000) W/m^2
Temperature coefficient of P_m	−0.3 %/°C to −0.5 %/°C
Temperature coefficient of I_m	+0.1 %/°C
Temperature coefficient of V_m	−0.38 %/°C
Temperature coefficient of Efficiency	−0.3 %/°C to −0.5 %/°C

4. **Negligible Impact of External Factors:** It is assumed that external factors (e.g., shading from nearby structures, dirt accumulation on panels) do not significantly affect the output power of the PV panels during the experiments, allowing for clear evaluation of the effects of positioning and area reduction.
5. **Steady State Operation:** The project assumes that the PV panels operate at a steady state during data collection, meaning that any transient effects (e.g., temporary variations in sunlight or shading) are minimized to ensure accurate measurements of power output.
6. **Efficiency of Interconnections:** It is assumed that the interconnection methods (e.g., Se-P connections) used in the study maintain high efficiency and do not introduce significant losses that could skew the results of power output measurements.
7. **Data Reliability:** The data obtained from experiments is assumed to be reliable and accurate, generated from calibrated instruments and measurement techniques that have been validated.
8. **Sufficient Sample Size:** It is assumed that the sample size of the PV panels and their configurations is sufficient to draw statistically significant conclusions regarding the optimal panel positioning and area reduction.

Fig. 2. PV panels placed at different positions: (a) VP PV, (b) FP PV, (c) IV SPV.

9. **Replicability of Results:** The project assumes that the experimental results can be replicated under similar conditions, indicating the robustness of the findings and their applicability to real-world scenarios.
10. **Economic Viability:** It is assumed that the optimization strategies and recommendations derived from the study will be economically viable and feasible for implementation in the solar energy market.

2.2. *Quantification of uncertainty*

- **Calibration Error:** Assume a calibration error of ±0.5 °C.
- **Environmental Influence**: Assume a standard deviation of ±1 °C due to ambient conditions.
- **Emissivity Variation:** Assume an uncertainty of ±1.5 °C due to variations in the emissivity of the PV panels.
- **Distance/Angle Variation:** Assume an additional uncertainty of ±0.5 °C due to the angle and distance from which measurements are taken [42].

3. Results and discussion

To validate the proposed study, the performance of 3 x 4 FP PV, VP PV, and IVP PV array is analyzed in terms of different interconnection schemes and at different variances of 62 cm, 77 cm and 92 cm. The performance of FP PV, VPPV and IVPPV are assessed in terms of output power and FF.

3.1. *Se-P interconnection*

In Se-P interconnection, the panels are connected in string to obtain the required voltage and these strings are connected in parallel to get the required current. The 12 panels are connected in Se-P interconnection to form a 3 x 4 PV array. The Se-P interconnections are commonly used in large-scale PV power generation.

3.1.1. *Se-P PV array mounted at a variance of 93 cm*

For the FP Se-P PV array, the PV array receives the highest incident irradiation of 1010 W/m^2 and generates the maximum power of 156 W with an FF of 0.67 at 11.30 a.m. as shown in Fig. 3 (a). Table 2 represents the minimum incident irradiation of 289 W/m^2 has produced the output power of 11 W at 06.00 p.m. In the FP Se-PV array, the average incident irradiation of 803 W/m^2 is recorded, which generated an average power of 84 W.

For the VP Se-P PV array, the maximum output power of 40 W is generated for the incident irradiation of 901 W/m^2 recorded at 11.30 a.m. with an FF of 0.67 as presented in Table 2. Similarly, the output power of 39 W is recorded for incident irradiation of 960 W/m^2 at 11.00 a. m as shown in Fig. 3 (b). The lowest output power of 6 W is produced for the irradiation of 260 W/m^{2}at 06.00 a.m. The average output power of 21 W for the average incident irradiation of 674 W/m^2.

Table 2 represents the maximum output power of 37 W for the incident irradiation of 812 W/m^2 recorded with an FF of 0.67 at 11.00 a.m. for the IVP Se-P PV array. Almost similarly output power of 36 W and 35 W is recorded for 820 W/m^2 and 750 W/m^2 at 11.30 a.m. and 10.30 a.m. At 06.00 a.m., the least output power of 6 W is recorded at an irradiation of 220 W/m^2 as shown in Fig. 3 (c). The average output power of 22 W for the average incident irradiation of 639 W/m^2.

3.1.2. *Se-P PV array mounted at a variance of 77 cm*

At 11.30 a.m., the maximum irradiation of 860 W/m^2 produced an output power of 106 W with an FF of 0.67 for the Se-P PV array as shown in Fig. 3 (d). This performance is followed by the output power of 102 W with irradiation of 750 W/m2 as reported in Table 3. For the irradiation of 220 W/m^2, the least output power of 10 W is produced at 06.00 p.m. The FPSe-P PV array has produced an average output power of 56 W for the average incident irradiation of 680 W/m^2.

For the VP Se-P PV array, the maximum output power of 40 W is generated with an FF of 0.67 at 11.30 a.m. for irradiation of 919 W/m^2 as shown in Fig. 3 (e). At 12.00 noon and 02.00 p.m., the PV array received irradiation of 1010 W/m^2 and 950 W/m^2 producing the output

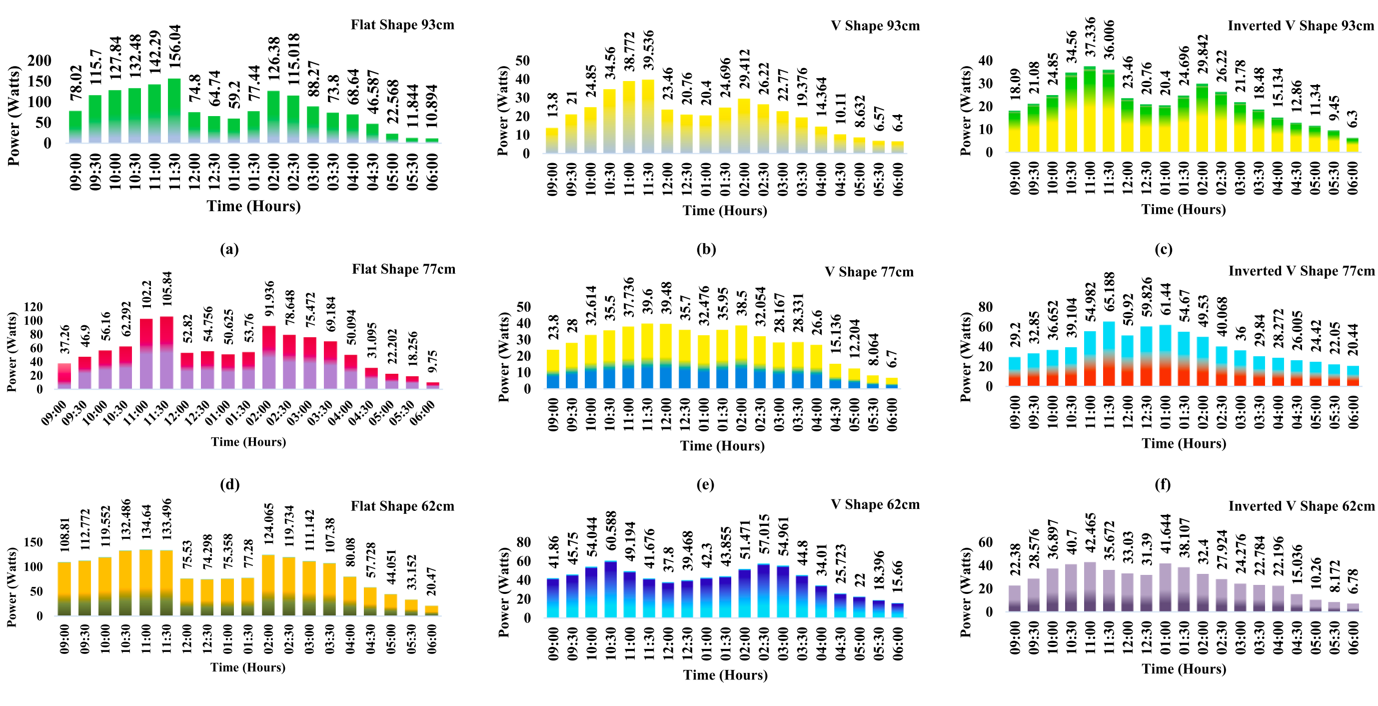

Fig. 3. Se-P PV array: (a) FP PV array at a variance of 93 cm, (b) VP PV array at a variance of 93 cm, (c) IVP PV array at a variance of 93 cm, (d) FP PV array at a variance of 77 cm, (e) VP PV array at a variance of 77 cm, (f) IVP PV array at a variance of 77 cm, (g) FP PV array at a variance of 62 cm, (h) VP PV array at a variance of 62 cm, (f) IVP PV array at a variance of 62 cm.

Table 2
Flat position PV array at 93 cm for Se-P interconnection.

Time (Hours)	FP PV Se-P array at 93 cm							VP PV Se-P array at 93 cm							IVP PV Se-P array at 93 cm						
	Irradiation (W/m^2)	V_{oc} (V)	I_{sc} (A)	V_m (V)	I_m (V)	P_m (W)	FF	Irradiation (W/m^2)	V_{oc} V)	I_{sc} (A)	V_m (V)	I_m (A)	P_m (W)	FF	Irradiation (W/m^2)	V_{oc} (V)	I_{sc} (A)	V_m (V)	I_m (V)	P_{max} (W)	FF
09:00	860	83.0	0.94	66.9	0.78	78	0.67	550	69.0	0.2	55.7	0.17	14	0.67	470	67.0	0.27	54.0	0.23	18	0.67
09:30	950	89.0	1.3	71.8	1.08	116	0.67	610	70.0	0.3	56.5	0.25	21	0.67	520	68.0	0.31	54.8	0.26	21	0.67
10:00	896	94.0	1.36	75.8	1.13	128	0.67	760	71.0	0.35	57.3	0.29	25	0.67	650	71.0	0.35	57.3	0.29	25	0.67
10:30	960	92.0	1.44	74.2	1.2	132	0.67	850	72.0	0.48	58.1	0.4	35	0.67	750	72.0	0.48	58.1	0.4	35	0.67
11:00	990	93.0	1.53	75.0	1.28	142	0.67	960	71.8	0.54	57.9	0.45	39	0.67	812	71.8	0.52	57.9	0.43	37	0.67
11:30	1010	94.0	1.66	75.8	1.38	156	0.67	901	70.6	0.56	56.9	0.47	40	0.67	820	70.6	0.51	56.9	0.43	36	0.67
12:00	970	88.0	0.85	71.0	0.71	75	0.67	870	69.0	0.34	55.7	0.28	23	0.67	830	69.0	0.34	55.7	0.28	23	0.67
12:30	980	78.0	0.83	62.9	0.69	65	0.67	990	69.2	0.3	55.8	0.25	21	0.67	890	69.2	0.3	55.8	0.25	21	0.67
01:00	950	74.0	0.8	59.7	0.67	59	0.67	860	68.0	0.3	54.8	0.25	20	0.67	860	68.0	0.3	54.8	0.25	20	0.67
01:30	950	88.0	0.88	71.0	0.73	77	0.67	810	68.6	0.36	55.3	0.3	25	0.67	870	68.6	0.36	55.3	0.3	25	0.67
02:00	910	89.0	1.42	71.8	1.18	126	0.67	840	68.4	0.43	55.2	0.36	29	0.67	840	69.4	0.43	56.0	0.36	30	0.67
02:30	970	87.8	1.31	70.8	1.09	115	0.67	716	69.0	0.38	55.7	0.32	26	0.67	716	69.0	0.38	55.7	0.32	26	0.67
03:00	821	91.0	0.97	73.4	0.81	88	0.67	620	69.0	0.33	55.7	0.28	23	0.67	670	66.0	0.33	53.2	0.28	22	0.67
03:30	740	90.0	0.82	72.6	0.68	74	0.67	580	69.2	0.28	55.8	0.23	19	0.67	610	66.0	0.28	53.2	0.23	18	0.67
04:00	672	88.0	0.78	71.0	0.65	69	0.67	490	68.4	0.21	55.2	0.18	14	0.67	520	65.8	0.23	53.1	0.19	15	0.67
04:30	560	87.9	0.53	70.9	0.44	47	0.67	440	67.4	0.15	54.4	0.13	10	0.67	450	64.3	0.2	51.9	0.17	13	0.67
05:00	440	86.8	0.26	70.0	0.22	23	0.67	380	66.4	0.13	53.6	0.11	9	0.67	350	63.0	0.18	50.8	0.15	11	0.67
05:30	330	84.6	0.14	68.2	0.12	12	0.67	320	65.7	0.1	53.0	0.08	7	0.67	290	63.0	0.15	50.8	0.13	9	0.67
06:00	289	83.8	0.13	67.6	0.11	11	0.67	260	64.0	0.1	51.6	0.08	6	0.67	220	63.0	0.1	50.8	0.08	6	0.67

Table 3
Flat position PV array at 77 cm for Se-P interconnection.

Time (Hours)	FP PV Se-P array at 77 cm							VP PV Se-P array at 77 cm							IVP PV Se-P array at 77 cm						
	Irradiation (W/m^2)	V_{oc} (V)	I_{sc} (A)	V_m (V)	I_m (V)	P_m (W)	FF	Irradiation (W/m^2)	V_{oc} V)	I_{sc} (A)	V_m (V)	I_m (A)	P_m (W)	FF	Irradiation (W/m^2)	V_{oc} (V)	I_{sc} (A)	V_m (V)	I_m (V)	P_{max} (W)	FF
09:00	470	69.0	0.54	55.7	0.45	37	0.67	490	68.0	0.35	54.8	0.29	24	0.67	590	73.0	0.40	58.8	0.33	29	0.67
09:30	530	70.0	0.67	56.5	0.56	47	0.67	589	70.0	0.4	56.5	0.33	28	0.67	650	73.0	0.45	58.8	0.38	33	0.67
10:00	650	72.0	0.78	58.1	0.65	56	0.67	632	70.9	0.46	57.2	0.38	33	0.67	760	74.8	0.49	60.3	0.41	37	0.67
10:30	730	71.6	0.87	57.7	0.73	62	0.67	720	71.0	0.5	57.3	0.42	36	0.67	810	75.2	0.52	60.6	0.43	39	0.67
11:00	750	73.0	1.4	58.9	1.17	102	0.67	850	71.2	0.53	57.4	0.44	38	0.67	850	74.3	0.74	59.9	0.62	55	0.67
11:30	860	72.0	1.47	58.1	1.23	106	0.67	919	72.0	0.55	58.1	0.46	40	0.67	930	75.8	0.86	61.1	0.72	65	0.67
12:00	850	69.5	0.76	56.1	0.63	53	0.67	1010	70.5	0.56	56.9	0.47	39	0.67	980	76.0	0.67	61.3	0.56	51	0.67
12:30	856	67.6	0.81	54.5	0.68	55	0.67	990	70.0	0.51	56.5	0.43	36	0.67	990	76.7	0.78	61.8	0.65	60	0.67
01:00	850	67.5	0.75	54.4	0.63	51	0.67	920	70.6	0.46	56.9	0.38	32	0.67	1005	76.8	0.80	61.9	0.67	61	0.67
01:30	774	67.2	0.8	54.2	0.67	54	0.67	1010	71.9	0.5	58.0	0.42	36	0.67	970	77.0	0.71	62.1	0.59	55	0.67
02:00	950	67.6	1.36	54.5	1.13	92	0.67	950	70.0	0.55	56.5	0.46	39	0.67	910	76.2	0.65	61.5	0.54	50	0.67
02:30	910	69.6	1.13	56.1	0.94	79	0.67	820	68.2	0.47	55.0	0.39	32	0.67	880	75.6	0.53	61.0	0.44	40	0.67
03:00	790	71.2	1.06	57.4	0.88	75	0.67	705	68.7	0.41	55.4	0.34	28	0.67	740	75.0	0.48	60.5	0.40	36	0.67
03:30	850	73.6	0.94	59.4	0.78	69	0.67	740	69.1	0.41	55.7	0.34	28	0.67	680	74.6	0.40	60.2	0.33	30	0.67
04:00	580	72.6	0.69	58.6	0.58	50	0.67	720	70.0	0.38	56.5	0.32	27	0.67	620	74.4	0.38	60.0	0.32	28	0.67
04:30	520	69.1	0.45	55.7	0.38	31	0.67	610	68.8	0.22	55.5	0.18	15	0.67	580	74.3	0.35	59.9	0.29	26	0.67
05:00	410	65.3	0.34	52.7	0.28	22	0.67	570	67.8	0.18	54.7	0.15	12	0.67	450	74.0	0.33	59.7	0.28	24	0.67
05:30	375	65.2	0.28	52.6	0.23	18	0.67	450	67.2	0.12	54.2	0.1	8	0.67	380	73.5	0.30	59.3	0.25	22	0.67
06:00	220	65.0	0.15	52.4	0.13	10	0.67	320	67.0	0.1	54.0	0.08	7	0.67	280	73.0	0.28	58.9	0.23	20	0.67

power of 39 W as presented in Table 3. The least output power of 7 W is recorded for the irradiation of 320 W/m^2 at 06.00 p.m. The average output power of 56 W is recorded for this case with an average incident irradiation of 680 W/m^2.

Fig. 3 (f) shows the IVP Se-P PV array has delivered the highest output power of 61 W with an FF of 0.67 for the incident irradiation of 1005 W/m^2 at 01.00 p.m. The almost similar output power of 60 W is delivered by the array for the incident irradiation of 990 W/m^2 at 12.30 p.m. as reported in Table 3. At 06.00 p.m., the array recorded the least power generation of 20 W for incident irradiation of 280 W/m^2. For the average incident irradiation of 740 W/m^2, the array has delivered an average output power of 40 W.

3.1.3. *Se-P PV array mounted at a variance of 62 cm*

The FP Se-P PV array has delivered the highest output power of 135 W with an FF of 0.67 for the incident irradiation of 850 W/m^2 at 11.00 a.m. as shown in Fig. 3 (g). At 11.30 a.m., the FP Se-P PV array generated almost similar output power of 133 W for the incident of 1010 W/m^2 as reported in Table 4. The lowest output power was generated for the incident irradiation of 250 W/m^2 at 06.00 p.m. The FP Se-P PV array has delivered an average output power of 92 W for the average incident irradiation of 250 W/m^2.

For the VP Se-P PV array, the maximum output power of 61 W is recorded for the incident irradiation of 840 W/m^2 with an FF of 0.67 at 10.30 a.m. as shown in Fig. 3 (h). Table 4 represents the lowest output power of 16 W produced for the incident irradiation of 298 W/m^2 at 06.00 p.m. For the average incident irradiation of 780 W/m^2, the VP Se-P PV array has generated an average output power of 41 W.

Fig. 3 (i) shows the IVP Se-P PV array, the maximum output power of 42 W is recorded for the incident irradiation of 890 W/m^2 & 947 W/m^2 with a FF of 0.67 at 11.00 a.m. and 01.00 p.m. The IVP Se-P PV array has recorded the least output power of 7 W for the incident irradiation of 290 W/m^2 at 06.00 p.m. in Table 4. The IVP Se-P PV array has generated the average output power of 27 W for the incident irradiation of 694 W/m^2.

3.1.4. *Inferences from Se-P connection*

At 93 cm, the flat-position (FP) Se-P PV array achieved a maximum power of 156 W for an irradiation of 1010 W/m^2, with an average power of 84 W. The vertical-position (VP) array recorded a maximum power of 40 W at 901 W/m^2 and an average power of 21 W. The inclined-position (IVP) array produced a peak power of 37 W with an average power of 22 W.

At 77 cm, the FP array produced 106 W at 860 W/m^2, with an average power of 56 W. The VP array generated a peak of 40 W at 919 W/m^2, and the IVP array delivered 61 W for 1005 W/m^2, with an average of 40 W.

At 62 cm, the FP array generated a maximum of 135 W at 850 W/m^2, with an average power of 92 W. The VP array achieved 61 W at 840 W/m^2, while the IVP array delivered a peak of 42 W with an average of 27 W.

3.2. *Total-cross-tied interconnection*

TCT interconnection in PV panels enhances energy yield by minimizing shading losses and mitigating panel mismatch, promoting uniform current distribution and reducing hot spots. TCT interconnection also enhances reliability by offering redundancy against partial shading or cell failures, thus optimizing overall system performance and longevity.

3.2.1. *TCT PV array mounted at a variance of 93 cm*

For the FPTCT PV array, the maximum output power of 123 W is measured for the incident irradiation of 920 W/m^2 with an FF of 0.67 at 11.30 a.m. as reported in Table 5. The almost similar output power of 122 W and 121 W is recorded for the irradiation of 960 W/m^2 and 820

Table 4
Flat position PV array at 62 cm for Se-P interconnection.

Time (Hours)	FPPV Se-P array at 62 cm							VPPV Se-P array at 62 cm							IVPPV Se-P array at 62 cm						
	Irradiation (W/m^2)	V_{oc} (V)	I_{sc} (A)	V_m (V)	I_m (V)	P_m (W)	FF	Irradiation (W/m^2)	V_{oc} V)	I_{sc} (A)	V_m (V)	I_m (A)	P_m (W)	FF	Irradiation (W/m^2)	V_{oc} (V)	I_{sc} (A)	V_m (V)	I_m (V)	P_{max} (W)	FF
09:00	476	93.0	1.17	75.0	0.98	109	0.67	510	91.0	0.46	73.4	0.38	42	0.67	560	74.6	0.30	60.2	0.25	22	0.67
09:30	580	93.2	1.21	75.2	1.01	113	0.67	650	91.5	0.50	73.8	0.42	46	0.67	620	75.2	0.38	60.7	0.32	29	0.67
10:00	650	93.4	1.28	75.3	1.07	120	0.67	734	91.6	0.59	73.9	0.49	54	0.67	750	75.3	0.49	60.7	0.41	37	0.67
10:30	790	93.3	1.42	75.2	1.18	132	0.67	840	91.8	0.66	74.0	0.55	61	0.67	830	74	0.55	59.7	0.46	41	0.67
11:00	850	93.5	1.44	75.4	1.20	135	0.67	910	91.1	0.54	73.5	0.45	49	0.67	890	74.5	0.57	60.1	0.48	42	0.67
11:30	1010	90.2	1.48	72.7	1.23	133	0.67	950	90.6	0.46	73.1	0.38	42	0.67	910	72.8	0.49	58.7	0.41	36	0.67
12:00	950	91.0	0.83	73.4	0.69	76	0.67	970	90.0	0.42	72.6	0.35	38	0.67	950	73.4	0.45	59.2	0.38	33	0.67
12:30	1070	91.5	0.81	73.8	0.68	74	0.67	990	89.7	0.44	72.3	0.37	39	0.67	1010	73	0.43	58.9	0.36	31	0.67
01:00	1040	91.9	0.82	74.1	0.68	75	0.67	1030	90.0	0.47	72.6	0.39	42	0.67	947	71.8	0.58	57.9	0.48	42	0.67
01:30	970	92.0	0.84	74.2	0.70	77	0.67	1010	89.5	0.49	72.2	0.41	44	0.67	910	71.9	0.53	58.0	0.44	38	0.67
02:00	960	91.9	1.35	74.1	1.13	124	0.67	820	90.3	0.57	72.8	0.48	51	0.67	870	72	0.45	58.1	0.38	32	0.67
02:30	955	91.4	1.31	73.7	1.09	120	0.67	940	90.5	0.63	73.0	0.53	57	0.67	750	71.6	0.39	57.7	0.33	28	0.67
03:00	920	91.1	1.22	73.5	1.02	111	0.67	940	90.1	0.61	72.7	0.51	55	0.67	630	71.4	0.34	57.6	0.28	24	0.67
03:30	770	91.0	1.18	73.4	0.98	107	0.67	820	89.6	0.50	72.3	0.42	45	0.67	590	71.2	0.32	57.4	0.27	23	0.67
04:00	690	91.0	0.88	73.4	0.73	80	0.67	680	89.5	0.38	72.2	0.32	34	0.67	550	71.6	0.31	57.7	0.26	22	0.67
04:30	650	90.2	0.64	72.7	0.53	58	0.67	700	88.7	0.29	71.5	0.24	26	0.67	430	71.6	0.21	57.7	0.18	15	0.67
05:00	570	89.9	0.49	72.5	0.41	44	0.67	560	88.0	0.25	71.0	0.21	22	0.67	380	68.4	0.15	55.2	0.13	10	0.67
05:30	380	89.6	0.37	72.3	0.31	33	0.67	470	87.6	0.21	70.7	0.18	18	0.67	320	68.1	0.12	54.9	0.10	8	0.67
06:00	250	89.0	0.23	71.8	0.19	20	0.67	298	87.0	0.18	70.2	0.15	16	0.67	290	67.8	0.10	54.7	0.08	7	0.67

Table 5
Flat position PV array at 93 cm for TCT configuration.

Time (Hours)	FPPV TCT array at 93 cm							VPPV TCT array at 93 cm							IVPPV TCT array at 93 cm						
	Irradiation (W/ m^2)	V_{oc} (V)	I_{sc} (A)	V_m (V)	I_m (V)	P_m (W)	FF	Irradiation (W/ m^2)	V_{oc} V)	I_{sc} (A)	V_m (V)	I_m (A)	P_m (W)	FF	Irradiation (W/ m^2)	V_{oc} (V)	I_{sc} (A)	V_m (V)	I_m (V)	P_{max} (W)	FF
09:00	650	92.6	0.6	74.7	0.50	56	0.67	500	72.1	0.20	58.2	0.17	14	0.67	470	69.7	0.27	56.2	0.23	19	0.67
09:30	720	93.4	0.8	75.3	0.67	75	0.67	570	72.9	0.40	58.8	0.33	29	0.67	580	70.5	0.30	56.9	0.25	21	0.67
10:00	760	93.6	0.9	75.5	0.75	84	0.67	610	73.1	0.50	59.0	0.42	37	0.67	680	70.7	0.39	57.0	0.33	28	0.67
10:30	820	94.2	1.0	76.0	0.83	94	0.67	670	73.7	0.60	59.4	0.50	44	0.67	750	71.3	0.44	57.5	0.37	31	0.67
11:00	860	94.6	1.2	76.3	1.00	114	0.67	710	74.1	0.80	59.8	0.67	59	0.67	810	71.7	0.51	57.8	0.43	37	0.67
11:30	920	94.7	1.3	76.4	1.08	123	0.67	810	74.2	0.90	59.8	0.75	67	0.67	900	71.8	0.62	57.9	0.52	45	0.67
12:00	960	94.3	1.3	76.1	1.08	122	0.67	930	73.8	0.89	59.5	0.74	66	0.67	925	71.4	0.58	57.6	0.48	41	0.67
12:30	1010	93.9	1.2	75.7	0.99	112	0.67	1060	73.4	0.79	59.2	0.66	58	0.67	1007	71	0.49	57.3	0.41	35	0.67
01:00	1100	94.0	1.0	75.8	0.88	99	0.67	950	73.5	0.65	59.3	0.54	48	0.67	942	71.1	0.52	57.3	0.43	37	0.67
01:30	990	94.2	1.1	76.0	0.92	104	0.67	840	73.7	0.70	59.4	0.58	52	0.67	870	71.3	0.44	57.5	0.37	31	0.67
02:00	910	93.7	1.1	75.6	0.98	111	0.67	760	73.2	0.78	59.0	0.65	57	0.67	840	70.8	0.41	57.1	0.34	29	0.67
02:30	890	94.3	1.2	76.1	1.00	113	0.67	740	73.8	0.80	59.5	0.67	59	0.67	714	71.4	0.37	57.6	0.31	26	0.67
03:00	860	94.5	1.2	76.2	1.03	117	0.67	710	74	0.84	59.7	0.70	62	0.67	670	71.6	0.35	57.7	0.29	25	0.67
03:30	820	94.0	1.2	75.8	1.08	121	0.67	670	73.5	0.89	59.3	0.74	65	0.67	615	71.1	0.31	57.3	0.26	22	0.67
04:00	760	93.8	1.2	75.7	1.01	113	0.67	610	73.3	0.81	59.1	0.68	59	0.67	523	70.9	0.26	57.2	0.22	18	0.67
04:30	720	93.5	1.1	75.4	0.98	110	0.67	570	73	0.78	58.9	0.65	57	0.67	450	70.6	0.22	56.9	0.18	16	0.67
05:00	680	93.2	1.0	75.2	0.83	93	0.67	530	72.7	0.60	58.6	0.50	44	0.67	360	70.3	0.19	56.7	0.16	13	0.67
05:30	630	93.0	0.9	75.0	0.75	84	0.67	510	72.5	0.50	58.5	0.42	36	0.67	290	70.1	0.15	56.5	0.13	11	0.67
06:00	590	92.8	0.8	74.8	0.67	74	0.67	440	72.3	0.40	58.3	0.33	29	0.67	220	69.9	0.12	56.4	0.10	8	0.67

W/m^2. For 650 W/m^2, the output power of 56 W is recorded at 09.00 p.m. as shown in Fig. 4 (a). The array has recorded the average output power of 101 W for the average incident irradiation of 824 W/m^2.

At 11.30 p.m., the maximum power of 67 W is generated by the VP TCT PV array for the incident irradiation of 810 W/m^2 with an FF of 0.67 as shown in Fig. 4 (b). Table 5 presents the least output power of 14 W generated for the irradiation of 500 W/m^2 at 09.00 a.m. For the average irradiation of 694 W/m^2, the average output power of 50 W is measured at 09.00 a.m.

In the IVP TCT PV array, the highest output power of 45 W is produced for the incident irradiation of 900 W/m^2 with an FF of 0.67 at 11.30 a.m. as reported in Table 5. This performance is followed by the output power of 41 W recorded for the irradiation of 590 W/m^2 at noon as shown in Fig. 4 (c). The average output power of 26 W is measured for the irradiation of 664 W/m^2.

3.2.2. TCT PV array mounted at a variance of 77 cm

In the FPTCT PV array, the incident irradiation of 940 W/m^2 delivered the output power of 102 W at noon as presented in Table 6. Almost, similar output power of 101 W is produced for the irradiation of 900 W/m^2 and 800 W/m^2 at 11.30 a.m. and 03.30 p.m. as shown in Fig. 4 (d). The output power of 39 W is recorded for irradiation of 630 W/m^2 at 09.00 a.m., which is the least for this case. The lowest output power of 82 W is recorded for the average irradiation of 824 W/m^2.

For the VP TCT PV array, the highest output power of 43 W is measured for the irradiation of 1020 W/m^2 at 01.00 noon as shown in Fig. 4 (e). For irradiation of 490 W/m^2, the least output power of 14 W is recorded at 09.00 a.m. with an FF of 0.67 as reported in Table 6. For the average irradiation of 763 W/m^2, the output power of 59 W is recorded for this case.

At noon, the maximum output power of 68 W is generated with an FF of 0.67 for the irradiation of 1010 W/m^2 by the IVP TCT PV array as shown in Fig. 4 (f). Table 6 presents the almost similar output power of 68 W produced at 12.30 p.m. for the irradiation of 800 W/m^2 at 12.30 p.m.

3.2.3. TCT PV array mounted at a variance of 62 cm

In the FPTCT PV array, the maximum output power of 131 W is generated for the irradiation of 1020 W/m^2 and 1070 W/m^2 at 11.30 a.m. and 12.30 p.m. as reported in Table 7. At noon, the output power of 129 W is produced by the array which is very close to the maximum output power as shown in Fig. 4 (g). The least output power of 51 W is produced for the irradiation of 480 W/m^2 with an FF of 0.67 at 09.00 a.m. At 09.00 a.m., the average output power of 86 W is generated for the average irradiation of 746 W/m^2.

For the incident irradiation of 880 W/m^2, the maximum output power of 45 W is generated with an FF of 0.67 at 02.30 p.m. by the VP TCT PV array as presented in Table 7. At 11.00 a.m., the output power of 44 W for the irradiation of 890 W/m^2 is measured as very close to the maximum output power as shown in Fig. 4 (h). For irradiation of 500 W/m^2, the lowest output power of 18 W is recorded at 09.00 a.m. The average output power of 33 W is produced for the average incident irradiation of 761 W/m^2.

In the IVP TCT PV array, the highest output power of 44 W is generated for the incident irradiation of 480 W/m^2 with an FF of 0.67 at 01.00 p.m. as presented in Table 7. The least output power of 10 W is recorded at 09.00 a.m. for the irradiation of 470 W/m^2 as shown in Fig. 4 (i). The average output power of 26 W is produced by the array for the average incident irradiation of 754 W/m^2.

The Total-Cross-Tied (TCT) interconnection enhances PV panel energy yield by reducing shading losses, promoting uniform current distribution, and mitigating hot spots. This setup also boosts system reliability by offering redundancy against partial shading or cell failures. Experimental testing was conducted at three different panel spacings: 93 cm, 77 cm, and 62 cm.

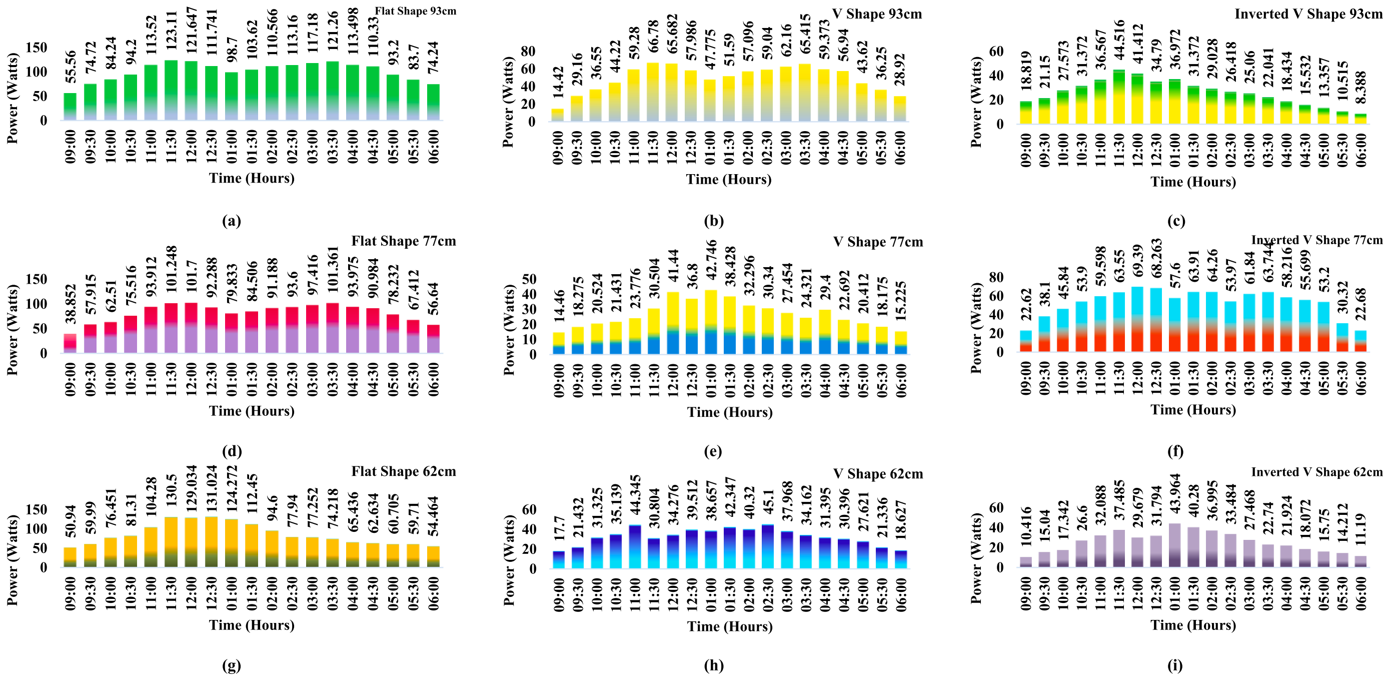

Fig. 4. TCT PV array: (a) FP PV array at a variance of 93 cm, (b) VP PV array at a variance of 93 cm, (c) IVP PV array at a variance of 93 cm, (d) FP PV array at a variance of 77 cm, (e) VP PV array at a variance of 77 cm, (f) IVP PV array at a variance of 77 cm, (g) FP PV array at a variance of 62 cm, (h) VP PV array at a variance of 62 cm, (f) IVP PV array at a variance of 62 cm.

3.2.4. Inferences from TCT connection

At 93 cm, the flat-position TCT PV array achieved a maximum output power of 123 W under an irradiation of 920 W/m^2, with a fill factor (FF) of 0.67. Similarly, for irradiations of 960 W/m^2 and 820 W/m^2, the output power remained around 122 W and 121 W, respectively. However, for lower irradiations of 650 W/m^2, the output power dropped to 56 W. The average power for this setup was 101 W with an average irradiation of 824 W/m^2.

For the 77 cm setup, a maximum output power of 102 W was measured at noon under 940 W/m^2 irradiation. Similar output power of 101 W was noted at 900 W/m^2 and 800 W/m^2, but at lower irradiations like 630 W/m^2, the power decreased to 39 W. The average power output was 82 W for an average irradiation of 824 W/m^2.

At 62 cm spacing, the TCT PV array recorded the highest output power of 131 W under 1020 W/m^2 and 1070 W/m^2 irradiation. The output power remained strong at 129 W around noon. However, for low irradiations like 480 W/m^2, the power dropped to 51 W. Overall, the average output power was 86 W for an average irradiation of 746 W/m^2. The findings highlight the synergies between interconnection strategy, panel positioning, and irradiance, maximizing power generation while reducing spatial footprint.

3.3. 2 Step Staircase interconnection

By utilizing the 2SS [16] method, solar PV systems can efficiently harness solar energy by optimizing both voltage and current levels. This method also allows for flexibility in system design, as it can accommodate varying numbers of panels and different voltage requirements. Additionally, it helps in minimizing power losses and maximizing the overall efficiency of the solar PV system. The performance of the 2SS [16] PV array is investigated for the different positions and mounted.

3.3.1. 2. Step Staircase [16] PV array mounted at a variance of 93 cm

At 12.30 noon, the maximum output power of 122 W is delivered by the FP 2SS [16] PV array for the incident irradiation of 990 W/m^2 as shown in Fig. 5 (a). Almost, similar output power of 120 W is produced for the irradiation of 910 W/m^2. The output power of 37 W is lowest produced by the array for irradiation of 560 W/m^2 with an FF of 0.67 at 09.00 a.m. as presented in Table 8. The array has produced an average output power of 73 W for the average incident irradiation of 759 W/m^2.

For the VP 2SS [16] PV array, the highest output power of 70 W is generated for the incident irradiation of 850 W/m^2 at 11.00 a.m. as reported in Table 8. At 02.30 p.m., the output power of 68 W is delivered by the VP 2SS [16] PV array which is close to the maximum power for the irradiation of 750 W/m^2 with a FF of 0.67 as shown in Fig. 5 (b). The lowest output power of 41 W is produced by the array for the irradiation of 250 W/m^2. For the average incident irradiation of 680 W/m^2, the array has produced an average output power of 57 W.

In the IVP 2SS [16] PV array, the maximum power of 74 W is delivered by the array for the incident irradiation of 998 W/m^2 with an FF of 0.67 at 02.00 p.m. as shown in Fig. 5 (c). The IVP 2SS [16] PV array has produced the least output power of 32 W for the irradiation of 480 W/m^2 at 09.00 a.m. as presented in Table 8. For this case, the average output power of 55 W is produced by the array for the average incident irradiation of 732 W/m^2.

3.3.2. 2 Step Staircase PV array mounted at a variance of 77 cm

For the FP 2SS [16] PV array, the output power of 77 W is generated for the incident irradiation of 978 W/m^2 with an FF of 0.67 at 02.30 p.m. as presented in Table 9. The output power of 29 W is the lowest produced by the array for the irradiation of 470 W/m^2 at 09.00 a.m. as shown in Fig. 5 (e). For the average incident irradiation of 770 W/m^2, the average output power of 57 W is delivered by the array.

Fig. 5 (f) shows the maximum output power of 73 W is produced by the VP 2SS [16] PV array for the incident irradiation of 879 W/m^2 & 755 W/m + at 12.00 noon and 03.30 p.m. Table 9 represents the lowest output power of 30 W is generated for the irradiation of 490 W/m^2 with a FF of 0.67 at 09.00 a.m. The array has produced the average output power of 56 W for the incident irradiation of 750 W/m^2 for this case.

For the IVP 2SS [16] PV array, the highest output power of 51 W for the incident irradiation of 921 W/m^2 with an FF of 0.67 at 02.30 p.m. as shown in Fig. 5 (f). At 11.00 a.m., the output power of 50 W is produced for the irradiation of 748 W/m^2, which is very close to the maximum power as represented in Table 9. For the average irradiation of 468

Table 6
Flat position PV array at 77 cm PV for TCT interconnection.

Time (Hours)	FPPV TCT array at 77 cm							VPPV TCT array at 77 cm							IVPPV TCT array at 77 cm						
	Irradiation (W/m^2)	V_{oc} (V)	I_{sc} (A)	V_m (V)	I_m (V)	P_m (W)	FF	Irradiation (W/m^2)	V_{oc} V)	I_{sc} (A)	V_m (V)	I_m (A)	P_m (W)	FF	Irradiation (W/m^2)	V_{oc} (V)	I_{sc} (A)	V_m (V)	I_m (V)	P_{max} (W)	FF
09:00	630	88.3	0.44	71.2	0.37	39	0.67	490	72.3	0.20	58.3	0.17	14	0.67	600	75.4	0.30	60.8	0.25	23	0.67
09:30	700	89.1	0.65	71.9	0.54	58	0.67	589	73.1	0.25	59.0	0.21	18	0.67	650	76.2	0.50	61.5	0.42	38	0.67
10:00	740	89.3	0.70	72.0	0.58	63	0.67	635	73.3	0.28	59.1	0.23	21	0.67	765	76.4	0.60	61.6	0.50	46	0.67
10:30	800	89.9	0.84	72.5	0.70	76	0.67	750	73.9	0.29	59.6	0.24	21	0.67	780	77	0.70	62.1	0.58	54	0.67
11:00	840	90.3	1.04	72.8	0.87	94	0.67	820	74.3	0.32	59.9	0.27	24	0.67	790	77.4	0.77	62.4	0.64	60	0.67
11:30	900	90.4	1.12	72.9	0.93	101	0.67	919	74.4	0.41	60.0	0.34	31	0.67	900	77.5	0.82	62.5	0.68	64	0.67
12:00	940	90	1.13	72.6	0.94	102	0.67	1010	74	0.56	59.7	0.47	41	0.67	1010	77.1	0.90	62.2	0.75	69	0.67
12:30	990	89.6	1.03	72.3	0.86	92	0.67	950	73.6	0.50	59.4	0.42	37	0.67	920	76.7	0.89	61.9	0.74	68	0.67
01:00	1080	89.7	0.89	72.3	0.74	80	0.67	1020	73.7	0.58	59.4	0.48	43	0.67	840	76.8	0.75	61.9	0.63	58	0.67
01:30	970	89.9	0.94	72.5	0.78	85	0.67	980	73.9	0.52	59.6	0.43	38	0.67	910	77	0.83	62.1	0.69	64	0.67
02:00	890	89.4	1.02	72.1	0.85	91	0.67	820	73.4	0.44	59.2	0.37	32	0.67	920	76.5	0.84	61.7	0.70	64	0.67
02:30	870	90	1.04	72.6	0.87	94	0.67	755	74	0.41	59.7	0.34	30	0.67	860	77.1	0.70	62.2	0.58	54	0.67
03:00	840	90.2	1.08	72.7	0.90	97	0.67	740	74.2	0.37	59.8	0.31	27	0.67	900	77.3	0.80	62.3	0.67	62	0.67
03:30	800	89.7	1.13	72.3	0.94	101	0.67	720	73.7	0.33	59.4	0.28	24	0.67	930	76.8	0.83	61.9	0.69	64	0.67
04:00	740	89.5	1.05	72.2	0.88	94	0.67	750	73.5	0.40	59.3	0.33	29	0.67	890	76.6	0.76	61.8	0.63	58	0.67
04:30	700	89.2	1.02	71.9	0.85	91	0.67	700	73.2	0.31	59.0	0.26	23	0.67	870	76.3	0.73	61.5	0.61	56	0.67
05:00	660	88.9	0.88	71.7	0.73	78	0.67	680	72.9	0.28	58.8	0.23	20	0.67	780	76	0.70	61.3	0.58	53	0.67
05:30	610	88.7	0.76	71.5	0.63	67	0.67	650	72.7	0.25	58.6	0.21	18	0.67	620	75.8	0.40	61.1	0.33	30	0.67
06:00	570	88.5	0.64	71.4	0.53	57	0.67	520	72.3	0.21	58.3	0.18	15	0.67	580	75.4	0.30	61.0	0.25	23	0.67

Table 7
Flat position PV array at 62 cm for TCT interconnection.

Time (Hours)	FP PV TCT array at 62 cm							VP PV TCT array at 62 cm							IV SPV TCT array at 62 cm						
	Irradiation (W/m^2)	V_{oc} (V)	I_{sc} (A)	V_m (V)	I_m (V)	P_m (W)	FF	Irradiation (W/m^2)	V_{oc} V)	I_{sc} (A)	V_m (V)	I_m (A)	P_m (W)	FF	Irradiation (W/m^2)	V_{oc} (V)	I_{sc} (A)	V_m (V)	I_m (V)	P_{max} (W)	FF
09:00	480	84.9	0.60	68.5	0.50	51	0.67	500	88.5	0.20	71.4	0.17	18	0.67	470	74.4	0.14	60.0	0.12	10	0.67
09:30	590	85.7	0.70	69.1	0.58	60	0.67	580	89.3	0.24	72.0	0.20	21	0.67	580	75.2	0.20	60.7	0.17	15	0.67
10:00	690	85.9	0.89	69.3	0.74	76	0.67	650	89.5	0.35	72.2	0.29	31	0.67	640	75.4	0.23	60.8	0.19	17	0.67
10:30	790	86.5	0.94	69.8	0.78	81	0.67	730	90.1	0.39	72.7	0.33	35	0.67	760	76.0	0.35	61.3	0.29	27	0.67
11:00	850	86.9	1.20	70.1	1.00	104	0.67	890	90.5	0.49	73.0	0.41	44	0.67	850	76.4	0.42	61.6	0.35	32	0.67
11:30	1020	87.0	1.50	70.2	1.25	131	0.67	950	90.6	0.34	73.1	0.28	31	0.67	990	76.5	0.49	61.7	0.41	37	0.67
12:00	950	86.6	1.49	69.8	1.24	129	0.67	1010	90.2	0.38	72.7	0.32	34	0.67	1000	76.1	0.39	61.4	0.33	30	0.67
12:30	1070	86.2	1.52	69.5	1.27	131	0.67	980	89.8	0.44	72.4	0.37	40	0.67	950	75.7	0.42	61.1	0.35	32	0.67
01:00	900	86.3	1.44	69.6	1.20	124	0.67	920	89.9	0.43	72.5	0.36	39	0.67	970	75.8	0.58	61.1	0.48	44	0.67
01:30	870	86.5	1.30	69.8	1.08	112	0.67	880	90.1	0.47	72.7	0.39	42	0.67	920	76.0	0.53	61.3	0.44	40	0.67
02:00	840	86.0	1.10	69.4	0.92	95	0.67	840	89.6	0.45	72.3	0.38	40	0.67	925	75.5	0.49	60.9	0.41	37	0.67
02:30	750	86.6	0.90	69.8	0.75	78	0.67	880	90.2	0.50	72.7	0.42	45	0.67	860	76.1	0.44	61.4	0.37	33	0.67
03:00	700	86.8	0.89	70.0	0.74	77	0.67	770	90.4	0.42	72.9	0.35	38	0.67	770	76.3	0.36	61.5	0.30	27	0.67
03:30	690	86.3	0.86	69.6	0.72	74	0.67	740	89.9	0.38	72.5	0.32	34	0.67	710	75.8	0.30	61.1	0.25	23	0.67
04:00	640	86.1	0.76	69.4	0.63	65	0.67	700	89.7	0.35	72.3	0.29	31	0.67	680	75.6	0.29	61.0	0.24	22	0.67
04:30	620	85.8	0.73	69.2	0.61	63	0.67	680	89.4	0.34	72.1	0.28	30	0.67	650	75.3	0.24	60.7	0.20	18	0.67
05:00	600	85.5	0.71	69.0	0.59	61	0.67	640	89.1	0.31	71.9	0.26	28	0.67	570	75.0	0.21	60.5	0.18	16	0.67
05:30	580	85.3	0.70	68.8	0.58	60	0.67	580	88.9	0.24	71.7	0.20	21	0.67	550	74.8	0.19	60.3	0.16	14	0.67
06:00	550	85.1	0.64	68.6	0.53	54	0.67	540	88.7	0.21	71.5	0.18	19	0.67	490	74.6	0.15	60.2	0.13	11	0.67

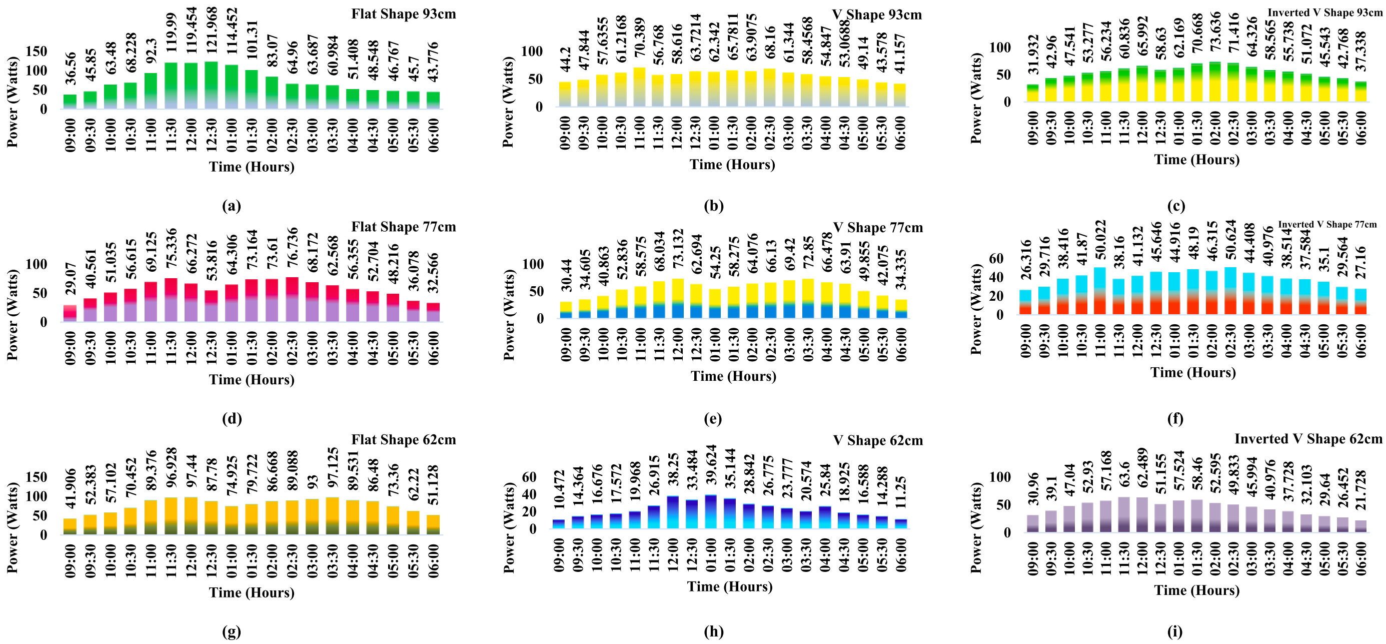

Fig. 5. 2SS PV array: (a) FP PV array at a variance of 93 cm, (b) VP PV array at a variance of 93 cm, (c) IVP PV array at a variance of 93 cm, (d) FP PV array at a variance of 77 cm, (e) VP PV array at a variance of 77 cm, (f) IVP PV array at a variance of 77 cm, (g) FP PV array at a variance of 62 cm, (h) VP PV array at a variance of 62 cm, (f) IVP PV array at a variance of 62 cm.

W/m^2, the output power of 26 W is produced by the array.

3.3.3. 2. Step Staircase PV array mounted at a variance of 62 cm

For the FP 2SS [16] PV array, the maximum output power of 97 W is generated for the incident irradiation of 845 W/m^2, 870 W/m^2, and 721 W/m2 at 11.30 a.m. as shown in Figs. 5 (g), 12.00 noon and 03.30 p.m. For irradiation of 489 W/m2, the least output power of 42 W is recorded with an FF of 0.67 at 09.00 a.m. as reported in Table 10. The average output power of 78 W is generated by the array for the average incident irradiation of 732 W/m^2.

In the VP 2SS [16] PV array, the highest output power of 40 W for the incident irradiation of 1006 W/m^2 at 01.00 p.m. as shown in Fig. 5 (h). For the incident irradiation of 567 W/m^2, the lowest output power of 10 W with an FF of 0.67 is represented in Table 10. In the VP 2SS [16] PV array, the average output power of 23 W is generated for the incident irradiation of 780 W/m^2.

For the incident irradiation of 921 W/m^2, the maximum power of 51 W is produced by the IVP 2SS [16] PV array with an FF of 0.67 at 02.30 p.m. as presented in Table 10. At 11.00 a.m., the output power of 50 W is generated for the irradiation of 748 W/m^2, which is very close to the maximum power as shown in Fig. 5 (i). The lowest output power of 26 W is produced by the array for the incident irradiation of 468 W/m^2. For the average incident irradiation of 753 W/m^2, the average output power of 40 W is generated by the array.

3.3.4. Inferences from Step Staircase (2SS) connection

For the FP 2SS PV array at a variance of 93 cm, it generated a maximum output power of 122 W at 990 W/m^2 around 12:30 p.m., with a similar output of 120 W for 910 W/m^2. The lowest power of 37 W was recorded for 560 W/m^2, while the average output was 73 W for 759 W/m^2.

The VP array at 93 cm produced a peak power of 70 W at 850 W/m^2 around 11:00 a.m., and at 2:30 p.m., it delivered 68 W for 750 W/m^2. The lowest power was 41 W for 250 W/m^2, and the average output was 57 W for 680 W/m^2.

The IVP array at 93 cm reached a maximum power of 74 W at 998 W/m^2, with an average output of 55 W for 732 W/m^2.

At 77 cm, the FP array generated a maximum of 77 W at 978 W/m^2, while the VP array produced 73 W at 879 W/m^2. The IVP array's highest output was 51 W at 921 W/m^2, with an average of 26 W for 468 W/m^2.

At 62 cm, the FP array delivered 97 W at 845 W/m^2, while the VP array reached 40 W at 1006 W/m^2. The IVP array's maximum output was 51 W at 921 W/m^2, and the average output was 40 W for 753 W/m^2.

3.3.5. Digest of the study

Contribution.

- This study presents an innovative approach to optimizing PV panel placement and interconnection strategies to maximize power output while minimizing spatial footprint.
- It experimentally evaluates flat, V-positioned, and inverted V-positioned PV arrays at different heights and interconnection schemes, providing novel insights into achieving higher efficiency in solar energy harvesting.

Merits.

- The findings directly contribute to advancing solar PV system designs, aligning with the journal's focus on innovative renewable energy solutions.
- This research not only enhances the understanding of spatial optimization for PV panels but also proposes practical configurations that can significantly impact the efficiency of solar farms and urban installations.

Specific Value.

- The study provides empirical evidence and engineering insights that benefit researchers, engineers, and policymakers working on sustainable and space-efficient solar energy solutions.
- The integration of novel stacking techniques and interconnection methods supports the global transition toward high-performance, land-efficient renewable energy systems.

Statement of Novelty:

This work introduces a structured experimental analysis of stacked

Table 8
Flat position PV array at 93 cm for 2 step staircase interconnection.

Time (Hours)	FP PV array at 93 cm							VP PV array at 93 cm							IV SPV array at 93 cm						
	Irradiation (W/m^2)	V_{oc} (V)	I_{sc} (A)	V_m (V)	I_m (V)	P_m (W)	FF	Irradiation (W/m^2)	V_{oc} V)	I_{sc} (A)	V_m (V)	I_m (A)	P_m (W)	FF	Irradiation (W/m^2)	V_{oc} (V)	I_{sc} (A)	V_m (V)	I_m (V)	P_{max} (W)	FF
09:00	560	91.4	0.40	73.7	0.33	37	0.67	520	88.4	0.50	71.3	0.42	44	0.67	480	88.7	0.36	71.5	0.30	32	0.67
09:30	645	91.7	0.50	74.0	0.42	46	0.67	580	88.6	0.54	71.5	0.45	48	0.67	547	89.5	0.48	72.2	0.40	43	0.67
10:00	776	92.0	0.69	74.2	0.58	63	0.67	640	88.6	0.65	71.5	0.54	58	0.67	634	89.7	0.53	72.3	0.44	48	0.67
10:30	820	92.2	0.74	74.4	0.62	68	0.67	745	88.7	0.69	71.6	0.58	61	0.67	698	90.3	0.59	72.8	0.49	53	0.67
11:00	868	92.3	1.00	74.4	0.83	92	0.67	850	89.1	0.79	71.9	0.66	70	0.67	717	90.7	0.62	73.1	0.52	56	0.67
11:30	910	92.3	1.30	74.4	1.08	120	0.67	890	88.7	0.64	71.5	0.53	57	0.67	780	90.8	0.67	73.2	0.56	61	0.67
12:00	946	92.6	1.29	74.7	1.08	119	0.67	960	86.2	0.68	69.5	0.57	59	0.67	868	90.4	0.73	72.9	0.61	66	0.67
12:30	990	92.4	1.32	74.5	1.10	122	0.67	980	86.1	0.74	69.4	0.62	64	0.67	960	90.2	0.65	72.7	0.54	59	0.67
01:00	1002	92.3	1.24	74.4	1.03	114	0.67	920	85.4	0.73	68.9	0.61	62	0.67	986	90.1	0.69	72.7	0.58	62	0.67
01:30	1010	92.1	1.10	74.3	0.92	101	0.67	890	85.4	0.77	68.9	0.64	66	0.67	1006	90.6	0.78	73.1	0.65	71	0.67
02:00	960	92.3	0.90	74.4	0.75	83	0.67	890	85.2	0.75	68.7	0.63	64	0.67	998	89.8	0.82	72.4	0.68	74	0.67
02:30	867	92.8	0.70	74.8	0.58	65	0.67	750	85.2	0.80	68.7	0.67	68	0.67	926	90.4	0.79	72.9	0.66	71	0.67
03:00	822	92.3	0.69	74.4	0.58	64	0.67	680	85.2	0.72	68.7	0.60	61	0.67	850	90.6	0.71	73.1	0.59	64	0.67
03:30	759	92.4	0.66	74.5	0.55	61	0.67	640	84.7	0.69	68.3	0.58	58	0.67	768	90.1	0.65	72.7	0.54	59	0.67
04:00	640	91.8	0.56	74.0	0.47	51	0.67	520	84.3	0.65	68.1	0.54	55	0.67	670	89.9	0.62	72.5	0.52	56	0.67
04:30	590	91.6	0.53	73.9	0.44	49	0.67	430	82.9	0.64	66.9	0.53	53	0.67	612	89.6	0.57	72.3	0.48	51	0.67
05:00	460	91.7	0.51	74.0	0.43	47	0.67	420	81.9	0.60	66.1	0.50	49	0.67	560	89.3	0.51	72.0	0.43	46	0.67
05:30	420	91.4	0.50	73.7	0.42	46	0.67	370	80.7	0.54	65.1	0.45	44	0.67	497	89.1	0.48	71.9	0.40	43	0.67
06:00	380	91.2	0.48	73.6	0.40	44	0.67	250	80.7	0.51	65.1	0.43	41	0.67	347	88.9	0.42	71.7	0.35	37	0.67

Table 9
Flat position PV array 77 cm for 2 step staircase interconnection.

Time (Hours)	FP PV array at 77 cm							VP PV array at 77 cm							IV SPV array at 77 cm						
	Irradiation (W/m^2)	V_{oc} (V)	I_{sc} (A)	V_m (V)	I_m (V)	P_m (W)	FF	Irradiation (W/m^2)	V_{oc} V)	I_{sc} (A)	V_m (V)	I_m (A)	P_m (W)	FF	Irradiation (W/m^2)	V_{oc} (V)	I_{sc} (A)	V_m (V)	I_m (V)	P_{max} (W)	FF
09:00	470	85.5	0.34	69.0	0.28	29	0.67	490	76.1	0.4	61.4	0.33	30	0.67	468	77.4	0.34	62.4	0.28	26	0.67
09:30	534	86.3	0.47	69.6	0.39	41	0.67	545	76.9	0.45	62.0	0.38	35	0.67	545	78.2	0.38	63.1	0.32	30	0.67
10:00	654	86.5	0.59	69.8	0.49	51	0.67	678	77.1	0.53	62.2	0.44	41	0.67	634	78.4	0.49	63.2	0.41	38	0.67
10:30	698	87.1	0.65	70.2	0.54	57	0.67	767	77.7	0.68	62.7	0.57	53	0.67	698	79.0	0.53	63.7	0.44	42	0.67
11:00	736	87.5	0.79	70.6	0.66	69	0.67	798	78.1	0.75	63.0	0.63	59	0.67	748	79.4	0.63	64.0	0.53	50	0.67
11:30	768	87.6	0.86	70.6	0.72	75	0.67	823	78.2	0.87	63.1	0.73	68	0.67	813	79.5	0.48	64.1	0.40	38	0.67
12:00	869	87.2	0.76	70.3	0.63	66	0.67	879	77.8	0.94	62.7	0.78	73	0.67	878	79.1	0.52	63.8	0.43	41	0.67
12:30	890	86.8	0.62	70.0	0.52	54	0.67	945	77.4	0.81	62.4	0.68	63	0.67	934	78.7	0.58	63.5	0.48	46	0.67
01:00	957	86.9	0.74	70.1	0.62	64	0.67	980	77.5	0.7	62.5	0.58	54	0.67	980	78.8	0.57	63.5	0.48	45	0.67
01:30	986	87.1	0.84	70.2	0.70	73	0.67	1090	77.7	0.75	62.7	0.63	58	0.67	1023	79.0	0.61	63.7	0.51	48	0.67
02:00	1020	86.6	0.85	69.8	0.71	74	0.67	935	77.2	0.83	62.3	0.69	64	0.67	967	78.5	0.59	63.3	0.49	46	0.67
02:30	978	87.2	0.88	70.3	0.73	77	0.67	912	77.8	0.85	62.7	0.71	66	0.67	921	79.1	0.64	63.8	0.53	51	0.67
03:00	928	87.4	0.78	70.5	0.65	68	0.67	870	78	0.89	62.9	0.74	69	0.67	878	79.3	0.56	64.0	0.47	44	0.67
03:30	879	86.9	0.72	70.1	0.60	63	0.67	755	77.5	0.94	62.5	0.78	73	0.67	756	78.8	0.52	63.5	0.43	41	0.67
04:00	823	86.7	0.65	69.9	0.54	56	0.67	697	77.3	0.86	62.3	0.72	66	0.67	712	78.6	0.49	63.4	0.41	39	0.67
04:30	780	86.4	0.61	69.7	0.51	53	0.67	634	77	0.83	62.1	0.69	64	0.67	687	78.3	0.48	63.1	0.40	38	0.67
05:00	654	86.1	0.56	69.4	0.47	48	0.67	578	76.7	0.65	61.9	0.54	50	0.67	642	78.0	0.45	62.9	0.38	35	0.67
05:30	547	85.9	0.42	69.3	0.35	36	0.67	498	76.5	0.55	61.7	0.46	42	0.67	589	77.8	0.38	62.7	0.32	30	0.67
06:00	467	85.7	0.38	69.1	0.32	33	0.67	369	76.3	0.45	61.5	0.38	34	0.67	443	77.6	0.35	62.6	0.29	27	0.67

Table 10
Flat position PV array at 62 cm for 2 step staircase interconnection.

Time (Hours)	FP PV array at 62 cm							VP PV array at 62 cm							IV SPV array at 62 cm						
	Irradiation (W/m^2)	V_{oc} (V)	I_{sc} (A)	V_m (V)	I_m (V)	P_m (W)	FF	Irradiation (W/m^2)	V_{oc} (V)	I_{sc} (A)	V_m (V)	I_m (A)	P_m (W)	FF	Irradiation (W/m^2)	V_{oc} (V)	I_{sc} (A)	V_m (V)	I_m (V)	P_{max} (W)	FF
09:00	489	91.1	0.46	73.5	0.38	42	0.67	567	74.8	0.14	60.3	0.12	10	0.67	468	77.4	0.34	62.4	0.28	26	0.67
09:30	547	91.9	0.57	74.1	0.48	52	0.67	678	75.6	0.19	61.0	0.16	14	0.67	545	78.2	0.38	63.1	0.32	30	0.67
10:00	598	92.1	0.62	74.3	0.52	57	0.67	723	75.8	0.22	61.1	0.18	17	0.67	634	78.4	0.49	63.2	0.41	38	0.67
10:30	678	92.7	0.76	74.8	0.63	70	0.67	789	76.4	0.23	61.6	0.19	18	0.67	698	79	0.53	63.7	0.44	42	0.67
11:00	789	93.1	0.96	75.1	0.80	89	0.67	823	76.8	0.26	61.9	0.22	20	0.67	748	79.4	0.63	64.0	0.53	50	0.67
11:30	845	93.2	1.04	75.2	0.87	97	0.67	897	76.9	0.35	62.0	0.29	27	0.67	813	79.5	0.48	64.1	0.40	38	0.67
12:00	870	92.8	1.05	74.8	0.88	97	0.67	945	76.5	0.5	61.7	0.42	38	0.67	878	79.1	0.52	63.8	0.43	41	0.67
12:30	956	92.4	0.95	74.5	0.79	88	0.67	989	76.1	0.44	61.4	0.37	33	0.67	934	78.7	0.58	63.5	0.48	46	0.67
01:00	989	92.5	0.81	74.6	0.68	75	0.67	1006	76.2	0.52	61.5	0.43	40	0.67	980	78.8	0.57	63.5	0.48	45	0.67
01:30	996	92.7	0.86	74.8	0.72	80	0.67	992	76.4	0.46	61.6	0.38	35	0.67	1023	79	0.61	63.7	0.51	48	0.67
02:00	888	92.2	0.94	74.4	0.78	87	0.67	912	75.9	0.38	61.2	0.32	29	0.67	967	78.5	0.59	63.3	0.49	46	0.67
02:30	848	92.8	0.96	74.8	0.80	89	0.67	878	76.5	0.35	61.7	0.29	27	0.67	921	79.1	0.64	63.8	0.53	51	0.67
03:00	789	93	1	75.0	0.83	93	0.67	812	76.7	0.31	61.9	0.26	24	0.67	878	79.3	0.56	64.0	0.47	44	0.67
03:30	721	92.5	1.05	74.6	0.88	97	0.67	768	76.2	0.27	61.5	0.23	21	0.67	756	78.8	0.52	63.5	0.43	41	0.67
04:00	698	92.3	0.97	74.4	0.81	90	0.67	712	76	0.34	61.3	0.28	26	0.67	712	78.6	0.49	63.4	0.41	39	0.67
04:30	632	92	0.94	74.2	0.78	86	0.67	654	75.7	0.25	61.0	0.21	19	0.67	687	78.3	0.48	63.1	0.40	38	0.67
05:00	580	91.7	0.8	74.0	0.67	73	0.67	623	75.4	0.22	60.8	0.18	17	0.67	642	78	0.45	62.9	0.38	35	0.67
05:30	521	91.5	0.68	73.8	0.57	62	0.67	589	75.2	0.19	60.6	0.16	14	0.67	589	77.8	0.38	62.7	0.32	30	0.67
06:00	479	91.3	0.56	73.6	0.47	51	0.67	456	75	0.15	60.5	0.13	11	0.67	443	77.6	0.35	62.6	0.29	27	0.67

PV panel configurations with varying interconnection schemes, optimizing power generation while reducing land usage. Unlike existing studies that focus primarily on single-plane PV arrays, this research systematically evaluates the performance of different stacking strategies under real-world conditions. The results highlight the superior efficiency of certain configurations, paving the way for future AI-driven optimizations in PV system design.

4. Conclusion

This study investigates the experimental verification for maximum the output power of different positioned PV array for reducing the panel area. The performance of different positioned PV array is investigated in terms of different interconnections and variance. From the experimental results, it is observed that the FP PV array has produced the superior output power compared to the VP and IVP PV positioned array. The Se-P PV array has produced the maximum output in comparison to TCT and 2SS. Further, for all the PV array has generated the highest output power for the variance of 93 cm followed by 62 cm and 77 cm.

A flat position PV array at 93 cm achieves the highest output power for Se-P interconnection of 156 W. A flat position PV array at 62 cm achieves the highest output power of 131 W for TCT interconnection. A flat position PV array at 93 cm achieves the highest output power of 122 W for 2 step staircase interconnections.

Thus, the experimental results show the significance of panel positioning to reduce the surface area of array. The research lays the groundwork for integrating future technologies like AI and advanced materials to further improve PV system performance. It can inform real-world solar energy applications, especially in urban and rural areas where space and efficiency are critical. This research would helpful the researchers, PV plant installers to enhance the output power with reduced land scape.

CRediT authorship contribution statement

Gurukarthik Babu Balachandran: Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Muthu Eshwaran Ramachandran:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Project administration, Conceptualization. **M. Palpandian:** Writing – review & editing, Writing – original draft, Validation, Software, Formal analysis, Conceptualization. **Prince Winston David:** Visualization, Project administration, Investigation, Funding acquisition, Conceptualization.

Statement of informed consent

There are no human subjects in this article and informed consent is not applicable.

Ethical approval

Ethical approval is not applicable for this article.

Statement of human and animal rights

This article does not contain any studies with human or animal subjects.

Declaration of funding

No funding was received.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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